On the Prediction of Non-Stationary Processes

Abdrabbo & Priestley (1967) — Detailed Technical Exposition

Authors: N. A. Abdrabbo and M. B. Priestley, University of Manchester
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Overview and Historical Context

This paper addresses one of the central open problems in mid-20th century stochastic process theory: how to extend the elegant Wiener–Kolmogorov prediction machinery from stationary processes to the vastly larger class of non-stationary processes. The Wiener–Kolmogorov framework — developed independently by Norbert Wiener and Andrei Kolmogorov in the early 1940s — had by 1967 become the canonical approach for linear least-squares prediction of stationary processes, with rigorous treatments available at varying levels of generality (Doob 1953, Bartlett 1955, Grenander and Rosenblatt 1957, Yaglom 1962, Whittle 1965). Its success rests on the existence of a spectral representation of stationary processes in terms of an orthogonal stochastic measure, which converts convolution operators into simple multiplicative ones in frequency space and reduces the prediction problem to a classical spectral factorization. [1]

The corresponding problem for non-stationary processes had, by 1967, received only fragmented attention. Parzen (1961) solved it in abstract Hilbert-space language, expressing the optimal predictor as an inner product but providing no operational formula. Cramér (1961a, 1961b) proved existence results without giving explicit predictor forms. Engineering-oriented treatments (Booton 1952, Zadeh 1953, Bendat 1956, 1957, 1959) reduced the problem to an integral equation for the predictor weights against the covariance function — equation (5.19) in the paper — but could only solve this equation under severely restrictive models. Kalman (1960) handled multivariate non-stationary processes governed by explicitly known linear differential equations, establishing the recursive filter that now bears his name, but requiring the dynamical equations as prior knowledge. Whittle (1965) treated autoregressive models with time-dependent coefficients and obtained explicit recursive predictors, results that the present paper partially recovers as special cases.[1]

The key obstacle common to all prior work was the absence of a canonical spectral representation for a broad class of non-stationary processes that would play the role that the Cramér–Kolmogorov spectral representation plays for stationary ones. This gap was precisely what Priestley's 1965 theory of *evolutionary spectra* filled, and the present paper exploits that theory to construct an almost exact analogue of the Wiener–Kolmogorov predictor for non-stationary processes.[2][3][1]

Section 2: Evolutionary Spectral Representations

The paper begins by summarizing the relevant parts of Priestley's (1965) theory. A zero-mean, finite-variance continuous-parameter process $\{X(t)\}$ is said to admit a *generalized spectral representation* if there exists a family $\mathscr{F} = \{\phi_t(\omega)\}$ of functions (indexed by time t) and a measure μ on $\mathbb{R}$ such that[3][1]

$$X(t) = \int_{-\infty}^{\infty} \phi_t(\omega)\, dZ(\omega), \tag{2.1}$$

where $\{Z(\omega)\}$ is an orthogonal process with $E|dZ(\omega)|^2 = \mu(d\omega)$[1]. The general class (2.1) was considered by Granger and Hatanaka (1964). The paper specializes to a subclass called *oscillatory processes*, defined by requiring that each $\phi_t(\omega)$ take the form

$$\phi_t(\omega) = e^{i\omega t} A_t(\omega), \tag{2.2}$$

where for each fixed ω the generalized Fourier transform of $A_t(\omega)$ (as a function of t) achieves an absolute maximum at the origin. The condition on the maximum ensures that the modulating envelope $A_t(\omega)$ varies slowly relative to the carrier $e^{i\omega t}$, giving the representation a physically meaningful time-frequency interpretation. For an oscillatory process with absolutely continuous μ (so that $\mu(d\omega) = f_0(\omega)\, d\omega$), the *evolutionary spectral density* at time t is defined by[1]

$$f_t(\omega) = |A_t(\omega)|^2 f_0(\omega). \tag{2.3}$$

This function is genuinely time-dependent: it describes the local distribution of power over frequency at epoch t, and collapses to the ordinary power spectral density when $\{X(t)\}$ is stationary (in which case $A_t(\omega) \equiv 1$). The discrete-parameter analogue is immediate: for $t \in \mathbb{Z}$,[2][3]

$$X(t) = \int_{-\pi}^{\pi} e^{i\omega t} A_t(\omega)\, dZ(\omega), \tag{2.4}$$

with $f_t(\omega) = |A_t(\omega)|^2 \mu(\{d\omega\})$ defined on $(-\pi, \pi]$[1]. The normalization convention $A_0(\omega) = 1$ for all ω is adopted throughout, so that $\mu(d\omega) = dF_0(\omega)$, the evolutionary spectrum at time zero[1].

Section 3: Moving-Average Representations for Oscillatory Processes

The engine of the Wiener–Kolmogorov theory is the *Wold decomposition*: every non-deterministic stationary process admits a one-sided moving-average (MA) representation driven by a white-noise innovation. Section 3 establishes an analogue for oscillatory processes — but with *time-dependent* MA coefficients — under suitable regularity conditions.[1]

3.1 The Discrete-Parameter Theorem

Let $\{X(t)\}$ be a discrete-parameter oscillatory process with absolutely continuous evolutionary spectrum $f_t(\omega) = |A_t(\omega)|^2 f(\omega)$, where $f(\omega) = f_0(\omega)$ is the base spectrum[1]. Two key conditions are imposed:

Condition C1 (regularity of base spectrum):

$$\int_{-\pi}^{\pi} \log f(\omega)\, d\omega > -\infty. \tag{3.2}$$

This is the classical Szegő–Kolmogorov condition ensuring that the base process is non-deterministic and admits a spectral factorization $|\psi(\omega)|^2 = f(\omega)$ with $\psi(z)$ analytic and zero-free inside the unit disc[1].

Condition C2 (analyticity of modulator): For each t, $A_t(z)$ has no poles or zeros inside the unit circle, so that each $A_t(\omega)$ has a causal (one-sided) Fourier series:

$$A_t(\omega) = \sum_{u=0}^{\infty} e^{-i\omega u} g_t(u). \tag{3.5}$$

A necessary condition for C2 is **Condition C3**:

$$\int_{-\pi}^{\pi} \log |A_t(\omega)|^2\, d\omega > -\infty \quad \text{for all } t. \tag{3.6}$$

Since both $\psi(\omega)$ and $A_t(\omega)$ are causal, their product $\alpha_t(\omega) = A_t(\omega)\psi(\omega)$ is also causal:

$$\alpha_t(\omega) = \sum_{u=0}^{\infty} e^{-i\omega u} h_t(u), \tag{3.9}$$

with $|\alpha_t(\omega)|^2 = f_t(\omega)$. The necessary condition for (3.9) is **Condition C4**:

$$\int_{-\pi}^{\pi} \log f_t(\omega)\, d\omega > -\infty \quad \text{for all } t, \tag{3.10}$$

and the paper shows $C_4 \Leftrightarrow C_1 \wedge C_3$. Introducing the auxiliary stationary uncorrelated process[1]

$$\xi(t) = \int_{-\pi}^{\pi} e^{i\omega t}\, dZ(\omega), \quad E\{\xi(t)\} = 0, \quad E\{|\xi(t)|^2\} = 2\pi, \quad E\{\xi(t)\xi^*(s)\} =$$

the main result of Section 3 is the following theorem:[1]

> **Theorem.** Let $\{X(t)\}$ be a discrete-parameter oscillatory process. If a family $\mathscr{F}$ satisfying C2 exists, and with respect to this family the evolutionary spectrum is absolutely continuous and satisfies C4, then $\{X(t)\}$ admits the one-sided MA representation
>
> $$X(t) = \sum_{u=0}^{\infty} h_t(u)\, \xi(t-u), \tag{3.13}$$
>
> where the series converges in mean square. Conversely, if (3.13) holds, then C4 must be satisfied.

The variance condition confirming mean-square convergence is:

$$2\pi \sum_{u=0}^{\infty} h_t^2(u) = \int_{-\pi}^{\pi} f_t(\omega)\, d\omega = \mathrm{var}\{X(t)\} < \infty. \tag{3.14}$$

The structural difference from the stationary Wold theorem is critical: in (3.13), the coefficient $h_t(u)$ depends on both the current time index t and the lag u, whereas in the stationary case the coefficient $h(u)$ depends on lag alone. The innovations $\{\xi(t)\}$ remain stationary and uncorrelated regardless of the non-stationarity of $\{X(t)\}$.[1]

The authors remark candidly that whereas in the stationary case C4 is *both necessary and sufficient* for the Wold representation, in the non-stationary case they can prove only necessity of C4 for the one-sided representation; sufficiency requires the additional analyticity Condition C2, which they cannot derive from C4 alone. This asymmetry has a counterpart in Cramér's (1961a) result that C4-type conditions for harmonizable processes imply *determinism* (absence of one-sided MA representations), underscoring that the non-stationary case is genuinely more delicate.[1]

3.2 The Continuous-Parameter Extension

For continuous-parameter oscillatory processes the argument proceeds analogously, with the discrete Szegő condition replaced by

$$(C_1^*) \qquad \int_{-\infty}^{\infty} \frac{\log f(\omega)}{1+\omega^2}\, d\omega > -\infty, \tag{3.15}$$

and C2 replaced by requiring that $A_t(\omega)$ have a causal Fourier integral representation

$$A_t(\omega) = \int_0^{\infty} e^{-i\omega u} g_t(u)\, du, \tag{3.16}$$

with corresponding condition C_3^* on $\log|A_t(\omega)|^2/(1+\omega^2)$[1]. The necessary condition on the evolutionary spectrum is

$$(C_4^*) \qquad \int_{-\infty}^{\infty} \frac{\log f_t(\omega)}{1+\omega^2}\, d\omega > -\infty \quad \text{for all } t. \tag{3.20}$$

The process is then representable as the stochastic convolution

$$X(t) = \int_0^{\infty} h_t(u)\, d\xi(t-u), \tag{3.23}$$

where $\{\xi(t)\}$ is formally the "derivative" of a Wiener process (i.e., a generalized white noise). The authors note that the continuous-parameter $\xi(t)$ requires careful distributional interpretation (cf. Whittle 1963, p. 21), which they handle by working with the Lebesgue–Stieltjes form (3.23) in place of the formally undefined pointwise integral.[1]

Section 4: Time-Domain Prediction

With the MA representation established, Section 4 derives the optimal linear predictor directly in the time domain — the analogue of the classical Wold predictor for stationary processes.[1]

4.1 Problem Formulation and Solution (Discrete)

The problem is: given $\{X(s), s \le t\}$, find the linear combination $\tilde{X}(t+m)$ minimizing the mean-square prediction error

$$M(m) = E\{|\tilde{X}(t+m) - X(t+m)|^2\}. \tag{4.1}$$

The existence and uniqueness of $\tilde{X}(t+m)$ follow from the Hilbert space projection theorem, exactly as in the stationary case. Granting representation (3.13), write[1]

$$X(t+m) = \underbrace{\sum_{u=m}^{\infty} h_{t+m}(u)\,\xi(t+m-u)}_{\text{past innovations}} + \underbrace{\sum_{u=0}^{m-1} h_{t+m}(u)\,\xi(t+m-u)}_{\text{future innovations}}. \tag{4.3}$$

Since the future innovations $\{\xi(t+1), \ldots, \xi(t+m)\}$ are orthogonal to the past $\{X(s), s \le t\} \equiv \{\xi(s), s \le t\}$, the projection is simply the "past" portion:

$$\tilde{X}(t+m) = \sum_{u=m}^{\infty} h_{t+m}(u)\,\xi(t+m-u) = \sum_{u=0}^{\infty} h_{t+m}(u+m)\,\xi(t-u). \tag{4.6}$$

The prediction variance is

$$M(m) = 2\pi \sum_{u=0}^{m-1} h_{t+m}^2(u). \tag{4.8}$$

Strikingly, the optimal predictor depends on $h_{t+m}(u)$ — the MA coefficients at the *future* time $t+m$, not the current time t. This is a fundamental and non-trivial feature of non-stationary prediction: in the stationary case $h_t \equiv h$ so the same coefficient function describes both present and future, but in the non-stationary case prediction at horizon m requires knowledge of the evolutionary spectrum at the future epoch $t+m$.[1]

To express $\tilde{X}(t+m)$ in terms of observable $\{X(s), s \le t\}$ rather than the latent $\{\xi(s)\}$, one must invert the MA representation. Write

$$\xi(t) = \sum_{v=0}^{\infty} k_t(v) X(t-v), \tag{4.9}$$

and substitute (3.13) into (4.9), giving the triangular system

$$\sum_{v=0}^{p} h_{t-v}(p-v)\, k_t(v) = \delta_{p,0}, \quad p = 0, 1, 2, \ldots, \text{ for all } t, \tag{4.10}$$

where $\delta_{p,0}$ is the Kronecker delta. The triangular structure means that the sequence $\{k_t(v)\}$ can be computed recursively by back-substitution:[1]

$$k_t(0) = 1/h_t(0), \quad k_t(1) = -h_t(1)/[h_t(0)\, h_{t-1}(0)], \quad \ldots$$

The dual system obtained by substituting (4.9) into (3.13) produces

$$\sum_{v=0}^{p} k_{t-v}(p-v)\, h_t(v) = \delta_{p,0}, \tag{4.11}$$

which is also triangular. Whittle (1965) had already noted that (4.10) and (4.11) are distinct in the non-stationary case but each implies the other, and the paper confirms this.[1]

4.2 Worked Examples (Time Domain)

Example 1 — Uniformly modulated process. Let $X(t) = C(t)Y(t)$ where $Y(t)$ is stationary with MA coefficients $g(u)$ and $C(t)$ is a deterministic envelope. The evolutionary spectrum is $f_t(\omega) = |C(t)|^2 f_y(\omega)$, so $h_t(u) = C(t)g(u)$[1]. The optimal predictor is

$$\tilde{X}(t+m) = C(t+m)\, \tilde{Y}(t+m),$$

where $\tilde{Y}(t+m)$ is the standard Wiener predictor of the underlying stationary process. More generally, if $X(t) = \sum_{u=0}^{\infty} a_t(u) Y(t-u)$, then $\tilde{X}(t+m) = \sum_{u=0}^{\infty} a_{t+m}(u) Y^*(t+m-u)$ where $Y^*(s) = Y(s)$ for $s \le t$ and $Y^*(s) = \tilde{Y}(s)$ for $s > t$. This result (noted by the referee) has a clean recursive flavor analogous to the stationary case.[1]

Example 2 — First-order autoregressive (AR(1)) with time-varying coefficients. The process satisfies

$$X(t) - \alpha(t) X(t-1) = \xi(t), \quad |\alpha(t)| < 1 \text{ for all } t. \tag{4.18}$$

The MA coefficients are

$$h_t(u) = \begin{cases} \alpha(t)\alpha(t-1)\cdots\alpha(t-u+1) & u > 0, \\ 1 & u = 0. \end{cases} \tag{4.19}$$

Applying formula (4.6) yields the exact predictor

$$\tilde{X}(t+m) = \alpha(t+m)\alpha(t+m-1)\cdots\alpha(t+1)\,X(t), \tag{4.20}$$

which is indeed the exact conditional expectation in the deterministic case $\xi \equiv 0$. The predictor chain-multiplies the m future autoregressive coefficients, a natural non-stationary generalization of $\hat{X}(t+m) = \alpha^m X(t)$ for the stationary AR(1).[1]

Example 3 — First-order moving-average (MA(1)) with time-varying coefficient. The process is $X(t) = \xi(t) - \alpha(t)\xi(t-1)$. The predictor is

$$\tilde{X}(t+m) = \begin{cases} -\alpha(t+1)\xi(t) & m = 1, \\ 0 & m > 1, \end{cases}$$

where $\xi(t) = \sum_{v=0}^{\infty} k_t(v)X(t-v)$ with $k_t(v) = \alpha(t)\alpha(t-1)\cdots\alpha(t-v+1)$. This mirrors the known stationary result — an MA(1) is only one-step predictable from its past, but the coefficient involves the *future* MA parameter $\alpha(t+1)$, not the current one, again reflecting the dependence on the spectrum at time $t+m$.[1]

Section 5: Frequency-Domain Prediction (Non-Stationary Wiener–Hopf)

Section 5 develops a frequency-domain approach that generalizes the Wiener–Hopf technique. Rather than working through innovations, it directly minimizes the spectral norm of the prediction error.[1]

5.1 Continuous-Parameter Formulation

Write the predictor as

$$\tilde{X}(t+m) = \int_0^{\infty} b_t(u)X(t-u)\,du, \tag{5.1}$$

where the weight function $b_t(u)$ may depend on both current time t and prediction horizon m. Using the evolutionary spectral representation, the predictor has the spectral form

$$\tilde{X}(t+m) = \int_{-\infty}^{\infty} e^{i\omega t} B_t(\omega)\,dZ(\omega), \tag{5.2}$$

where

$$B_t(\omega) = \int_0^{\infty} b_t(u)\,\alpha_{t-u}(\omega)e^{-i\omega u}\,du. \tag{5.3}$$

The mean-square prediction error is therefore

$$M(m) = \int_{-\infty}^{\infty} |e^{i\omega m}\alpha_{t+m}(\omega) - B_t(\omega)|^2\,d\omega. \tag{5.4}$$

The paper shows that $B_t(\omega)$ is necessarily a *backward* (causal) transform — i.e., its inverse Fourier transform $K_t(v)$ vanishes for $v < 0$:[1]

$$K_t(v) = \int_0^v b_t(u)\, h_{t-u}(v-u)\, du. \tag{5.6}$$

The key step is to decompose the target $e^{i\omega m}\alpha_{t+m}(\omega)$ into its backward (causal) part $C_t^{(1)}(\omega)$ and forward (anti-causal) part $C_t^{(2)}(\omega)$. Since the backward and forward parts are orthogonal in $L^2(d\omega)$, the error (5.4) decomposes as

$$M(m) = \int_{-\infty}^{\infty} |C_t^{(1)}(\omega) - B_t(\omega)|^2\, d\omega + \int_{-\infty}^{\infty} |C_t^{(2)}(\omega)|^2\, d\omega. \tag{5.7}$$

The first term is minimized to zero by setting $B_t(\omega) = C_t^{(1)}(\omega)$, and the irreducible prediction error is

$$M(m) = \int_{-\infty}^{\infty} |C_t^{(2)}(\omega)|^2\, d\omega. \tag{5.9}$$

Taking Fourier transforms of the optimality condition $B_t(\omega) = C_t^{(1)}(\omega)$ and using (5.6) yields the *master integral equation* for the optimal weight function:

$$\int_0^v b_t(u)\, h_{t-u}(v-u)\, du = h_{t+m}(v+m), \quad v \geq 0,\ \text{all } t. \tag{5.10}$$

This is a Volterra integral equation of the first kind in b_t. In the stationary case $h_{t-u}(v-u) \equiv h(v-u)$ and $h_{t+m}(v+m) \equiv h(v+m)$, so (5.10) becomes a convolution equation solvable immediately by taking the ratio of Fourier transforms — the classical Wiener–Hopf approach. In the non-stationary case the kernel depends on two time arguments and no convolution structure is available, so a different approach is required.[1]

5.2 Converting to a Volterra Equation of the Second Kind

Under the additional assumptions that $h_{t-v}(0) \neq 0$ for all $v > 0$ and that $h_{t-u}(v-u)$ is differentiable in v for $u \leq v$, differentiating (5.10) with respect to v converts it to a Volterra equation of the second kind:

$$b_t(v) + \int_0^v \frac{h'_{t-u}(v-u)}{h_{t-v}(0)} b_t(u)\, du = \frac{h'_{t+m}(v+m)}{h_{t-v}(0)}, \quad v \geq 0, \tag{5.11}$$

where $h'_{t-u}(v-u) = \partial h_{t-u}(v-u)/\partial v$. The solution to (5.11) is given via the resolvent kernel (Tricomi 1957):[1]

$$b_t(v) = \frac{h'_{t+m}(v+m)}{h_{t-v}(0)} + \int_0^v H_t^*(v,u,-1) \frac{h'_{t+m}(u+m)}{h_{t-u}(0)}\, du, \tag{5.12}$$

where the resolvent $H_t^*(v,u,-1) = -\sum_{p=0}^{\infty} (-1)^p K^*_{t,p+1}(u,v)$ is built from iterated kernels

$$K^*_{t,p+1}(u,v) = \int_u^v K^*_{t,1}(u,z)\, K^*_{t,p}(z,v)\, dz, \quad K^*_{t,1}(u,v) = \frac{h'_{t-u}(v-u)}{h_{t-v}(0)}.$$

Equation (5.12) provides a formal closed-form expression for $\{b_t(u)\}$, though the authors note that numerical solution of (5.10) directly is often more practical.[1]

5.3 Discrete-Parameter Frequency-Domain Results

For discrete-parameter processes, the predictor $\tilde{X}(t+m) = \sum_{u=0}^{\infty} b_t(u) X(t-u)$ leads, by a parallel argument, to the discrete analogue of the master equation:

$$\sum_{u=0}^{v} b_t(u)\, h_{t-u}(v-u) = h_{t+m}(v+m), \quad v = 0, 1, 2, \dots \tag{5.18}$$

This is again a triangular (lower-triangular in matrix form) system, solvable step by step by back-substitution. This triangular structure, identical in form to the systems (4.10) and (4.11) of Section 4, is identified by the authors as the key computational advantage of the spectral approach: the covariance-based system (5.19) has no such structure and is intractable in general, while the spectral approach reduces it to triangular form automatically.[1]

The authors report that this procedure was implemented numerically on an ATLAS computer and proved successful.[1]

5.4 Frequency-Domain Verification of Examples

Example 4 (AR(1)). Substituting $h_t(u)$ from (4.19) into (5.18) and solving the triangular system confirms:

$$b_t(0) = \alpha(t+m)\cdots\alpha(t+1), \quad b_t(u) = 0 \text{ for } u > 0,$$

recovering $\tilde{X}(t+m) = \alpha(t+m)\cdots\alpha(t+1)\, X(t)$ in agreement with the time-domain result.[1]

Example 5 (MA(1)). With $h_t(0) = 1, h_t(1) = -\alpha(t), h_t(u) = 0$ for $u > 1$, solving (5.18) gives, for all $u \geq 0$:

$$b_t(u) = \begin{cases} \alpha(t+1)\alpha(t)\cdots\alpha(t-u+1) & m = 1, \\ 0 & m > 1, \end{cases}$$

recovering the time-domain predictor of Example 3 and confirming internal consistency of the two approaches.[1]

Section 6: Discussion and Practical Implications

The discussion section addresses the gap between the theoretical framework and practical application. Throughout the paper, the second-order structure (i.e., the evolutionary spectra $\{f_t(\omega)\}$ for all t) is assumed known. In practice, the predictor $\tilde{X}(t+m)$ is determined by $h_{t+m}(u)$, the MA coefficients at the *future* time $t+m$, which requires knowledge of $f_{t+m}(\omega)$ — a quantity not directly observable from data available up to time t.[1]

The authors identify two practically relevant regimes. When the prediction step m is much smaller than the characteristic width B_x of the process (a bandwidth parameter measuring the rate of spectral evolution, defined in Priestley 1965), the approximation $f_{t+m}(\omega) \approx f_t(\omega)$ is reasonable — the evolutionary spectrum changes so slowly that the most recently estimated spectrum is a good proxy for the future one. When m is comparable to or larger than B_x, however, genuine long-range extrapolation of the spectral structure is required. The authors suggest two strategies: (i) parametric extrapolation, if the model specifies how the time-varying parameters $\{\alpha_j(t)\}$ evolve (e.g., as periodic functions of t), or (ii) regression-based extrapolation on the spectral ordinates themselves when no parametric form is assumed.[1]

The authors conclude with the important qualitative observation that the very nature of non-stationarity *precludes* reliable long-range prediction unless strong structural assumptions are made about how the non-stationarity evolves. This is not a limitation of their method but an intrinsic feature of non-stationary inference: without regularity constraints on the evolutionary spectrum, any predictor for large m would be extrapolating into an unconstrained function space.[1]

Structural Comparison: Stationary vs. Non-Stationary Prediction

Feature	Stationary (Wiener–Kolmogorov)	Non-Stationary (Abdrabbo–Priestley)
Spectral representation	$X(t) = \int e^{i\omega t} dZ(\omega)$	$X(t) = \int e^{i\omega t} A_t(\omega)\, dZ(\omega)$
Spectrum	Fixed $f(\omega)$	Time-varying $f_t(\omega) = \lvert A_t(\omega)\rvert^2 f(\omega)$
MA coefficients	$h(u)$, lag-only	$h_t(u)$, time-and-lag dependent
Wold-type represent	One-sided MA; C4 necessary **and** sufficient	One-sided MA; C4 necessary, C2+C4 sufficient

Feature	**Stationary (Wiener–Kolmogorov)**	**Non-Stationary (Abdrabbo–Priestley)**
ation		
Optimal predictor form	$\tilde{X}(t+m) = \sum_{u \geq m} h(u)\xi(t+m-u)$	$\tilde{X}(t+m) = \sum_{u \geq m} h_{t+m}(u)\xi(t+m-u)$
Predictor coefficients depend on	Current time (trivially, since stationary)	**Future** time $t+m$

Feature	Stationary (Wiener–Kolmogorov)	Non-Stationary (Abdrabbo–Priestley)
Master equation	Convolution (Wiener–Hopf)	Volterra first-kind (5.10) / second-kind (5.11)
Key computational structure	Spectral division	Triangular back-substitution
Long-range predict	Extrapolation of fixed spectrum	Requires model for spectral evolution

Feature	Stationary (Wiener–Kolmogorov)	Non-Stationary (Abdrabbo–Priestley)
ion		

Mathematical Significance and Legacy

The paper's primary contribution is the demonstration that the Wiener–Kolmogorov prediction theory extends, without fundamental obstruction, to the class of oscillatory processes via the evolutionary spectral framework. The extension is not merely formal: it yields operationally explicit predictors and prediction-error formulae in terms of quantities (the MA coefficient sequences $\{h_t(u)\}$) that are in principle computable from the evolutionary spectral density.[2][1]

Several features distinguish the paper technically. The identification of the triangular structure in (4.10), (4.11), and (5.18) as the key to computational tractability is a central insight: it transforms an otherwise ill-posed integral equation (5.19) into a sequence of one-dimensional linear equations solvable by back-substitution. The derivation of the prediction-error formula (5.9) as the L^2 norm of the forward (anti-causal) component of the target's spectral factor has an elegant frequency-domain interpretation — the irreducible error is precisely the energy in the "future" part of the target that cannot be explained by past data — paralleling the classical derivation but generalized to time-varying spectral factors.[1]

The dependence of the optimal predictor on the *future* evolutionary spectrum $f_{t+m}(\omega)$ is a genuinely new phenomenon with no counterpart in stationary theory, and the paper is clear-eyed about its practical consequences. Methods for estimating evolutionary spectra from single realizations were developed in Priestley (1965, 1966), making the framework empirically applicable in principle, though the additional complication of spectrum estimation is deferred. This theoretical apparatus subsequently became foundational for time-varying spectral analysis, with applications ranging from structural engineering (non-stationary seismic ground motion models) to biomedical signal processing (EEG time-frequency analysis).[4][5][6][3][2][1]

The connection to Whittle (1965) is explicitly acknowledged: the paper's results for autoregressive models with time-varying coefficients recover Whittle's recursive predictors as special cases of the general framework. This positions the paper as both a generalization and a unification — subsuming the existing special-case results within a single spectral-theoretic structure, and providing a systematic approach for cases (e.g., moving-average models, uniformly modulated processes) not covered by Whittle's recursions.[1]

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